same in all directions. The relativity principle of Lorentz, Poincaré, and Einstein differs from that of Galileo and Newton, but it has this same implication: the physics as perceived by an observer in uniform motion is indeed identical with that perceived by an observer at rest.

Yet, in the new relativity, Maxwell's equations *are* consistent with this principle, and the speed of light is measured to have a definite fixed value in every direction, no matter in what direction or with what speed the observer might be moving. How is this magic achieved so that these apparently hopelessly incompatible requirements are reconciled? I shall leave it to Feynman to explain—in his own inimitable fashion.

Relativity is perhaps the first place where the physical power of the mathematical idea of *symmetry* begins to be felt. Symmetry is a familiar idea, but it is less familiar to people how such an idea can be applied in accordance with a set of mathematical expressions. But it is just such a thing that is needed in order to implement the principles of special relativity in a system of equations. In order to be consistent with the relativity principle, whereby physics "looks the same" to an observer in uniform motion as to an observer at rest, there must be a "symmetry transformation" which translates one observer's measured quantities into those of the other. It is a symmetry because the physical laws appear the same to each observer, and "symmetry," after all, asserts that something has the same appearance from two distinct points of view. Feynman's approach to abstract matters of this nature is very down to earth, and he is able to convey the ideas in a way that is accessible to people with no particular mathematical experience or aptitude for abstract thinking.

Whereas relativity pointed the way to additional symmetries that had not been perceived before, some of the more modern developments in physics have shown that certain symmetries, previously thought to be universal, are in fact subtly violated. It came as one of the most profound shocks to the physical community in 1957, as the work of Lee, Yang, and Wu showed, that in certain basic